

Department of MCA

MCA Sem: Ist Sec: B-1, B-2 & B-3

Session 2024-25 (Odd Semester)

Assignment-2

Subject Name: Problem Solving using C

Subject Code: BMC102

Date of Assignment: 11/12/2024

Date of Submission: 14/12/2024

UNIT -2

Conditional Program Execution: if, if-else, and nested if-else statements, Switch statements, Restrictions on switch values, Use of break and default with switch, Comparison of switch and if-else.

Loops and Iteration: for, while and do-while loops, Multiple loop variables, Nested loops, Assignment operators, break and continue statement.

Functions: Introduction, Types, Declaration of a Function, Function calls, defining functions, Function Prototypes, passing arguments to a function Return values and their types, writing multifunction program, Calling function by value, Recursive functions.

Q. No.	Questions	Marks	CO	KL
1	Examine the statement “Decision making refers to the process of evaluating alternatives and selecting a course of action based on the analysis of information, estimation of probabilities, and assigning values to anticipated outcomes” and deduce the method(s) to apply decision making in C language. Also, apply the “decision making” methods for solving a simple problem of “going to college from home” and showcase all the alternatives using a flowchart/program.	5	CO4	K4
2	Make use of a situation, where a Software program in a mobile device allows user to unlock the mobile with 5 password attempts and after that it resets mobile device, to explain the concept of iterations in a computer program.	2	CO4	K3
3	Examine the image below and explain how does this particular task relates to which specific control structure(s) in C-Language. 	2	CO4	K4
4	Illustrate the terms “Reusability”, “Modularity”, “Optimization” with reference to usage of function(s) in a computer program. Also Build a program having a function for calculation of second highest value in an array. The array needs to be passed to the function and the	2	CO4	K3

	function should return the second highest value which is to be displayed in main function.			
5	Build a program in C-Language to convert a decimal number into its equivalent binary and octal form using recursive functions.	2	CO4	K3

Signature & Name of Faculty
Date:

HOD Signature
Date:

CO-Course Outcomes mapped with respective question KL-
Bloom's Knowledge Level (K1, K2, K3, K4, K5, and K6)

K1 – Remember K2 – Understand K3 – Apply K4 – Analyze K5 – Evaluate K6- Create

Note: Use A-4 size sheets to write their answer. Avoid copying assignments